

Feature Mining and Detection of AI-Generated Text

Haibin Lai, Shan Zhang, Yitao Zheng, Chenjun Xu, Taojie Wang*

Southern University of Science and Technology

CS306 Data Mining Course Project, Group 6

Shenzhen, China

{12211612,12213052,12210830,12210126,12210519}@mail.sustech.edu.cn

Abstract

Detecting AI-generated text is increasingly difficult because modern large language models (LLMs) differ in generation era, model family, decoding strategy, and writing domain. We present an interpretable feature-based detector organized around a two-layer taxonomy: 36 model-free statistical features and 82 reference-model features from embeddings, LLM’s perplexity, and token probabilities. On RAID—a benchmark spanning 11 generators, 8 domains, and 11 adversarial attacks—our detector reaches AUC 0.9992 with 118 combined features, far surpassing zero-shot Fast-DetectGPT (AUC 0.7815) while remaining fully explainable through SHAP. Our further finding is that detection features have shelf lives, which no single feature group works everywhere. We also find noise dilution that naively concatenating many weak features can lower AUC. Together these results argue that practical detectors should fuse interpretable feature groups only after measuring each group’s target-domain signal strength, rather than assuming more features are always better.

Keywords

AI-generated text detection, interpretable machine learning, feature engineering, SHAP, adversarial robustness

1 Introduction

Large language models such as ChatGPT, GPT-4, Claude, Llama, and Mistral can generate fluent text at scale. This creates risks for academic integrity, misinformation, and content provenance. A detector used in practice should satisfy three requirements: it should perform well across multiple generators, remain robust across writing domains and adversarial attacks, and explain why a document was classified as AI-generated.

Existing detectors fall into three broad paradigms. Supervised neural classifiers, such as fine-tuned RoBERTa [18] models, can achieve high in-distribution accuracy but are difficult to interpret and often degrade under distribution shift. Zero-shot detectors such as DetectGPT [20], Fast-DetectGPT [1], and Binoculars [13] avoid labeled training data, but their probability assumptions depend heavily on the reference model. Feature-based detectors are lightweight and interpretable, but many prior studies evaluate them on single-generator datasets such as HC3 [10], leaving open whether they remain useful in multi-generator benchmarks such as RAID [8].

Our project follows the defense storyline: we do not only ask whether a detector is accurate; we ask *which feature families work, when they expire, and how to combine them without adding noise*. The central claim is that no single reference model or feature group dominates all settings. Instead, feature groups have effective ranges determined by the target generator era and model family.

Contributions. We make three contributions:

- (1) We implement an interpretable 118-dimensional feature pipeline: 36 model-free statistics plus 82 reference-model features from BERT, GPT-2, and Mistral-7B.
- (2) We achieve AUC 0.9992 on RAID using XGBoost, with SHAP explanations showing how paragraph structure and token-probability signals jointly drive predictions.
- (3) We provide a cross-dataset analysis revealing two phenomena that motivate target-domain feature selection: feature shelf life, where each feature family fails on a different generator regime; and noise dilution, where adding many weak features hurts performance.

2 Method

2.1 Problem Formulation and Pipeline

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ denote a collection of documents, where x_i is a text and $y_i \in \{0, 1\}$ indicates human-written ($y_i = 0$) or AI-generated ($y_i = 1$) content. Our detector maps each document to a heterogeneous feature vector

$$\mathbf{z}(x) = [\mathbf{s}(x); \mathbf{e}(x); \mathbf{p}(x); \mathbf{t}(x)], \quad (1)$$

where $\mathbf{s} \in \mathbb{R}^{36}$ contains model-free statistics, $\mathbf{e} \in \mathbb{R}^{50}$ is a PCA-compressed sentence embedding, $\mathbf{p} \in \mathbb{R}^2$ contains GPT-2 likelihood features, and $\mathbf{t} \in \mathbb{R}^{30}$ summarizes token probabilities under an observer language model. The experimental feature matrix therefore contains 118 columns. An XGBoost classifier learns $\hat{p}_i = P(y_i = 1 | \mathbf{z}(x_i))$.

The pipeline has four stages: (1) deterministic text parsing, (2) parallel extraction of model-free and reference-model features, (3) feature-level quality control and optional group diagnostics, and (4) supervised classification followed by SHAP-based explanation, as illustrated in Figure 1. All learned transforms, including PCA and feature scaling, are fitted on the training partition only.

In the parsing stage, the model-free extractor operates on the original text without paraphrasing or semantic normalization, since formatting is itself a useful signal. Words are separated by whitespace, sentences are split at sequences of ., !, and ?, and paragraphs are separated by blank lines. For the neural feature extractors, documents are tokenized with the corresponding pretrained tokenizer and truncated to at most 512 subword tokens. Empty or invalid numerical outputs are treated as missing values and imputed using statistics estimated from the training set.

2.2 Model-Free Linguistic Features

The model-free layer describes what the document looks like without querying a pretrained neural model. Its design principle is that AI-generated text leaves stylistic and structural fingerprints in the *surface form* of a document—how long paragraphs are, how evenly sentences are sized, how diverse the vocabulary is, and how punctuation is distributed—even when its content is fluent and topically appropriate. Because none of these statistics requires a reference model, they are cheap to compute, deterministic, language-agnostic in spirit, and fully

*Student IDs: Haibin Lai 12211612, Shan Zhang 12213052, Yitao Zheng 12210830, Chenjun Xu 12210126, Taojie Wang 12210519.

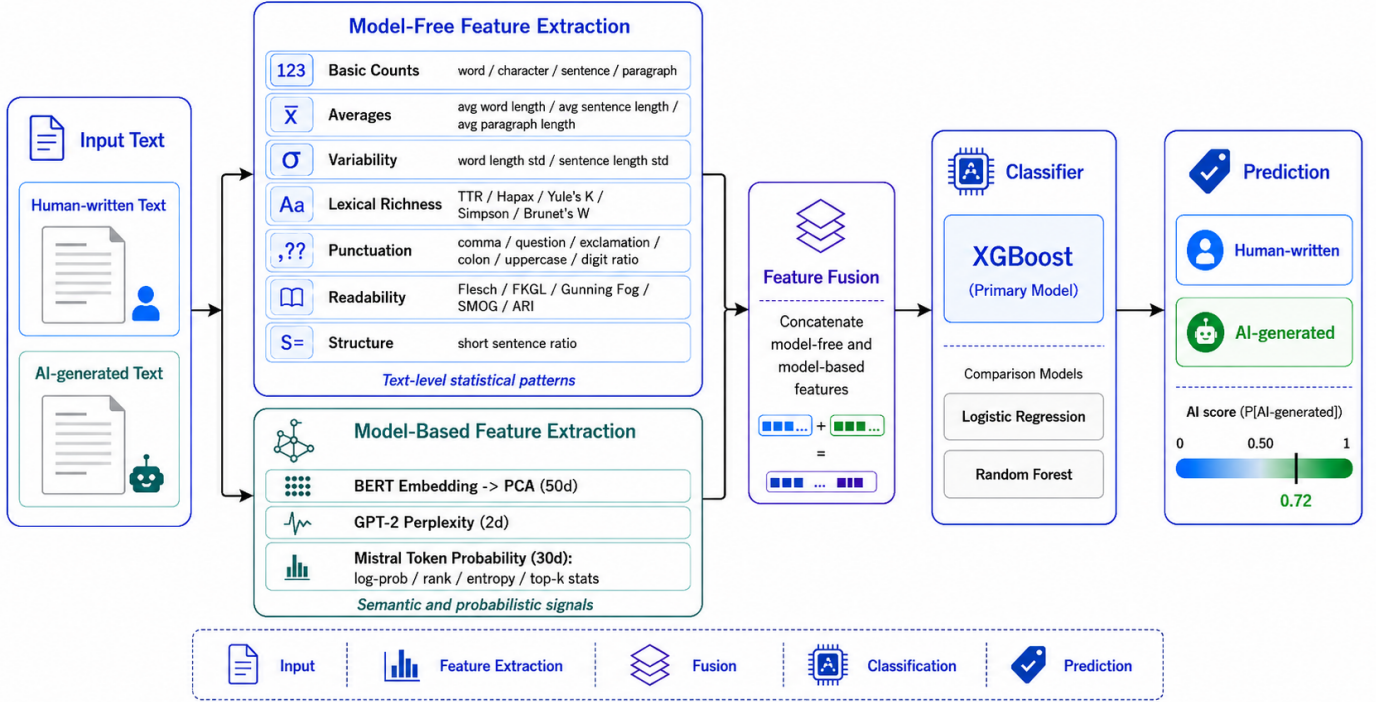


Figure 1: Overall framework. Each document is described by a model-free feature extractor (basic counts, averages, variability, lexical richness, punctuation, readability, and structure) and a model-based extractor (BERT embedding compressed by PCA, GPT-2 perplexity, and Mistral token-probability statistics). The two feature groups are concatenated and classified by XGBoost, with Logistic Regression and Random Forest as comparison models, producing an AI-generation probability.

interpretable: every value corresponds to a named, human-readable property of the text. The layer contains seven groups, summarized in Table 1.

Table 1: Model-free feature groups used by the detector.

Group	Dim.	Representative features
basic counts	4	words, characters, sentences, paragraphs
averages	4	word length, sentence length, paragraph length
variability	2	word- and sentence-length standard deviation
lexical richness	7	TTR, hapax ratio, Yule’s K, Simpson index
punctuation	11	punctuation, uppercase, digit, whitespace rates
readability	7	Flesch, Fog, SMOG, ARI, Dale–Chall
structure	1	fraction of sentences shorter than five words

Counts, averages, and variability. The first three groups capture document scale and its regularity. Basic counts (words, characters, sentences, paragraphs) record raw size; averages (mean word, sentence, and paragraph length) normalize that size into style descriptors; and variability (the standard deviation of word and sentence lengths) measures *rhythm*. The intuition is that human writing tends to alternate between short and long sentences, whereas many LLMs emit prose with a more uniform cadence and predictable paragraph segmentation. As the cross-dataset and SHAP analyses later confirm, these structural descriptors turn out to be the strongest model-free

signals on RAID, but here we introduce them purely on linguistic grounds.

Lexical richness. The lexical-richness group quantifies vocabulary diversity using classical stylometric measures [29]. Let N be the number of tokens, V the number of distinct lower-cased tokens, V_1 the number of tokens occurring exactly once, and f_j the frequency of vocabulary item j . Representative measures are

$$\text{TTR} = \frac{V}{N}, \quad \text{Hapax} = \frac{V_1}{N}, \quad (2)$$

$$\text{YuleK} = 10^4 \frac{\sum_j f_j^2 - N}{N^2}, \quad \text{Simpson} = 1 - \frac{\sum_j f_j(f_j - 1)}{N(N - 1)}. \quad (3)$$

The type–token ratio (TTR) and hapax ratio measure how often new and unique words appear, while Yule’s K [30] and the Simpson index [27] measure how concentrated the frequency distribution is; both are comparatively length-robust because they aggregate over the full frequency spectrum rather than a raw count. These features test the hypothesis that LLM decoding, especially under low-temperature or nucleus sampling, can produce slightly more repetitive or more uniformly diverse vocabulary than human authors.

Punctuation and orthography. The punctuation group records the per-character rates of commas, periods, exclamation marks, question marks, semicolons, colons, quotes, and dashes, together with the uppercase, digit, and whitespace ratios. These are stylistic and formatting fingerprints: list-like or heavily segmented AI output, distinctive quoting conventions, and consistent capitalization all surface here

without any semantic analysis. Because they operate on characters rather than tokens, they also act as a tripwire for character-level adversarial edits.

Readability. The readability group applies seven standard indices—Flesch Reading Ease [9], Flesch–Kincaid grade [15], Gunning Fog [12], SMOG [19], Coleman–Liau [4], the Automated Readability Index (ARI) [28], and Dale–Chall [6]—each combining sentence length and word or syllable complexity into a grade-level estimate. They summarize how syntactically and lexically demanding the text is. All seven are computed with the same implementation for every dataset so that scores are directly comparable across benchmarks.

Structure. Finally, a single structural feature records the fraction of sentences shorter than five words, a compact proxy for fragmentary or list-style writing.

Taken together, count features capture document scale while normalized rates capture style. Paragraph-level measures are emphasized because, as the SHAP analysis in Section 4.5 confirms, they carry most of the model-free discriminative power: LLMs frequently produce regularly sized, highly segmented prose with low rhythm variation, whereas human writing shows more irregular paragraph length and sentence structure. Because these signals are aggregate statistics rather than exact strings, they are also relatively robust to lexical paraphrase and synonym substitution, a property we revisit in the adversarial-robustness analysis.

2.3 Reference-Model Features

The reference-model layer complements the surface statistics by probing how plausible a document looks to an external neural model. The three groups are deliberately complementary: BERT embeddings capture semantic and stylistic representation, GPT-2 perplexity scores the overall likelihood under a single reference distribution, and Mistral-7B token statistics describe the local, per-token probability structure.

2.3.1 Sentence Embedding. We encode each document with Sentence-Transformers model `all-MiniLM-L6-v2` [22], a BERT-based [7] encoder, producing a 384-dimensional sentence embedding. PCA is fitted on training embeddings and retains the first 50 components:

$$\mathbf{e}(\mathbf{x}) = \mathbf{W}_{50}^T (\mathbf{h}(\mathbf{x}) - \boldsymbol{\mu}_{\text{train}}). \quad (4)$$

These components provide a compact representation of semantic and stylistic variation. Unlike the named statistical features, individual PCA components are latent; consequently, we describe the overall detector as feature-based and post-hoc explainable rather than fully intrinsically interpretable.

2.3.2 GPT-2 Likelihood. For a token sequence $w_{1:T}$, GPT-2 [21] gives the mean negative log-likelihood

$$\text{NLL}(\mathbf{x}) = -\frac{1}{T-1} \sum_{t=2}^T \log P_{\text{GPT2}}(w_t \mid w_{<t}). \quad (5)$$

Perplexity is computed separately for each document as

$$\text{PPL}(\mathbf{x}) = \exp(\text{NLL}(\mathbf{x})). \quad (6)$$

The two retained features are $\text{PPL}(\mathbf{x})$ and $\log(1 + \text{PPL}(\mathbf{x}))$. These features should be interpreted as compatibility with the GPT-2 distribution, not as universal evidence of AI generation.

2.3.3 Observer-Model Token Statistics. Mistral-7B-Instruct-v0.1 [14] serves as an observer model. At every position t , we compute the log-probability of the observed token,

$$\ell_t = \log P_{\text{obs}}(w_t \mid w_{<t}), \quad (7)$$

its rank r_t among vocabulary logits, the predictive entropy

$$H_t = - \sum_v P_{\text{obs}}(v \mid w_{<t}) \log P_{\text{obs}}(v \mid w_{<t}), \quad (8)$$

and the cumulative probability of the top-1 and top-5 predictions.

The 30 document-level features comprise nine log-probability statistics (mean, standard deviation, minimum, maximum, median, 10th and 90th percentiles, skewness, and kurtosis), six entropy statistics, eight rank statistics, four top- k probability statistics, two adjacent log-probability-difference statistics, and sequence length. The rank features include the fractions of observed tokens appearing in the observer’s top-1, top-5, top-10, and top-100 predictions. These aggregates measure how predictable and locally smooth the observed token sequence appears to the observer model, generalizing the token-rank visualization of GLTR [11] into document-level statistics. Intuitively, AI-generated text tends to receive higher log-probabilities, lower entropy, and top-ranked tokens more often than human text, which is less predictable to the observer.

2.4 Feature Fusion and Classifiers

All three classifiers we evaluate are interpretable (white-box) models trained on the same feature matrix. The primary model is **XG-Boost** [3] with 500 trees, maximum depth 8, learning rate 0.1, row subsampling 0.8, and column subsampling 0.8. Gradient-boosted trees can capture interactions between macro-level structure and micro-level token probabilities without requiring the feature groups to share a common scale, and their split structure together with SHAP yields per-prediction attribution.

We additionally train two secondary white-box classifiers on the identical features so that the contribution of the feature design can be separated from that of any single learner. **Random Forest** [2] uses 300 trees with maximum depth 15. **Logistic Regression** [5] is L_2 -regularized with $C = 1.0$ and a linear decision boundary whose standardized coefficients are directly readable; its continuous inputs are standardized using training-set means and variances. All classifiers use a fixed random seed and, where the class distribution is imbalanced, balanced class weighting.

The full 118-column model is the primary reported configuration. We also train models on individual groups and selected combinations to distinguish independent signal from complementary signal.

2.5 Group Signal Diagnostics

For diagnostic analysis, the direction-free strength of feature j is

$$a_j = \max\{\text{AUC}(z_j, y), 1 - \text{AUC}(z_j, y)\}, \quad (9)$$

and the score of group g is

$$S_g = \frac{1}{|g|} \sum_{j \in g} a_j. \quad (10)$$

All scores are computed from training data only. A value near 0.5 indicates that the group has little marginal signal in the target dataset. However, this statistic does not measure nonlinear interactions and

Table 2: Datasets used for evaluation.

Dataset	Role	Generators	Domains	Size
RAID	primary	11	8	468K
HC3	auxiliary	1	4–5	85K
SemEval 2024	auxiliary	6+	multi	154K
TuringBench	auxiliary	19	news	332K
Pile	auxiliary	mixed	multi	80K sample

therefore is used as a diagnostic rather than as a guarantee that removing a group will improve test performance. Any deployed group-selection rule should be chosen inside cross-validation and compared against the unfiltered model on an untouched test set.

2.6 Explanation

We apply SHAP TreeExplainer [16, 17] to the trained XGBoost model. Global importance is measured by mean absolute SHAP value, while instance-level waterfall plots show which features move a prediction toward the AI or human class. We also report single-feature AUC and group-only ablations. These views answer different questions: single-feature AUC measures marginal discrimination, group ablation measures joint predictive value, and SHAP describes how the final model uses a feature in context.

3 Experimental Setup

3.1 Datasets

Table 2 lists the five datasets. RAID [8] is the primary benchmark because it spans generator families, domains, decoding strategies, and adversarial attacks. HC3 [10] tests single-generator ChatGPT detection. SemEval 2024 [24] represents modern multi-generator detection, where competitive systems such as Genaios [23] combine language-model probabilistic features similar to ours. TuringBench [25] contains older GPT-1/2/3-era generators, and Pile [26] provides mixed academic-style text.

3.2 Evaluation Protocol and Metrics

For the reported within-dataset experiments, we use a fixed random seed of 42 and stratified train/test partitions. RAID is subsampled to at most 50,000 documents per class and then split 80/20, producing at most 80,000 training and 20,000 test documents. Feature extraction is performed after the split; PCA, scaling, imputation, and any feature-group selection are fitted using training data only.

Because random row splits can place texts derived from the same prompt or source document in both partitions, a stronger evaluation should additionally use group-disjoint splits based on prompt or source identifiers. We recommend reporting three complementary settings:

- (1) **In-distribution:** stratified split within each dataset.
- (2) **Generator/domain shift:** leave-one-generator-out and leave-one-domain-out evaluation on RAID.
- (3) **Cross-dataset transfer:** train on one benchmark and test on a different benchmark without refitting the classifier.

Results obtained by independently training and testing within five datasets should be called *multi-dataset evaluation*, not cross-dataset generalization.

ROC AUC is the primary metric because it is threshold-independent. Accuracy, precision, recall, and F1 are reported at a threshold selected

Table 3: Method comparison on RAID.

Method	AUC	Accuracy
XGBoost, 118 combined	0.9992	0.9912
XGBoost, 88 stat+model	0.9951	0.9714
XGBoost, 30 token	0.9900	0.9590
Random Forest, 118 combined	0.9981	0.9859
Logistic Regression, 118 combined	0.9935	0.9706
Fast-DetectGPT, zero-shot	0.7815	0.7452

on training or validation data and then fixed before test evaluation. For the final report, 95% bootstrap confidence intervals should accompany AUC differences, especially when comparing configurations whose scores differ only in the third decimal place.

4 Results

4.1 Main RAID Results

Table 3 shows the main RAID comparison. The 118-feature XGBoost detector reaches AUC 0.9992 and accuracy 0.9912, improving over the 88-feature baseline and over the 30-token-only model. It substantially outperforms Fast-DetectGPT, demonstrating that a supervised but interpretable feature pipeline can be more reliable than a generic zero-shot probability-curvature score in a multi-generator benchmark.

4.2 Single-Feature Discrimination

Before combining feature groups, we examine which individual features separate human from AI text on RAID. Figure 2 shows the distributions of the strongest model-free features. Paragraph-level structure dominates: `words_per_paragraph` is the single most discriminative feature (AUC 0.89), with `sentences_per_paragraph` and `paragraph_count` close behind, whereas character-level rates such as `uppercase_ratio` overlap heavily between the two classes. The gap is consistent with the qualitative observation that AI generators tend to produce regularly sized, highly segmented paragraphs, while human writing varies more.

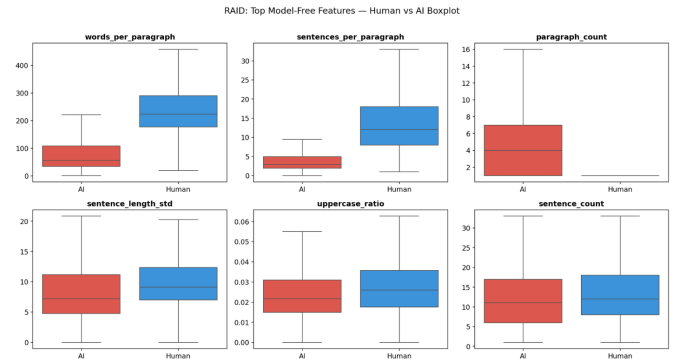


Figure 2: Human vs. AI distributions of the strongest model-free features on RAID. Paragraph-structure features separate the classes most cleanly, while character-level rates overlap.

The observer-model token features show a complementary pattern (Figure 3). Under Mistral, AI text is more *predictable*: it has higher mean log-probability (`lp_mean`), more tokens ranked in the top-100 (`rank_top100_frac`), and lower local variability (`lp_std`,

lp_diff_std) than human text. These token statistics capture micro-level probability structure that the model-free features cannot, motivating their combination.

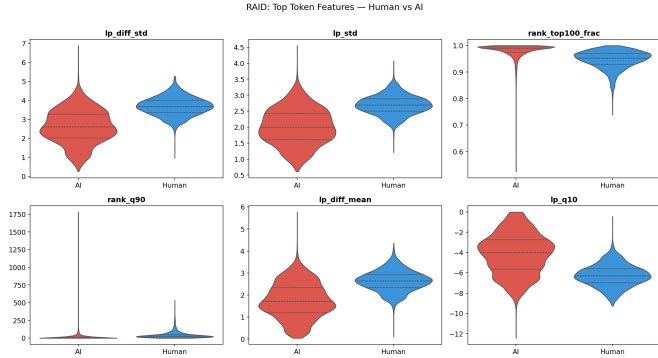


Figure 3: Human vs. AI distributions of the strongest observer-model (Mistral) token features on RAID. AI text is more predictable: higher log-probability, higher top-rank fraction, and lower local variability.

4.3 Feature Shelf Life Across Datasets

Table 4 is the core cross-dataset result. Each feature group has a different effective range. GPT-2 perplexity is excellent for GPT-family text, achieving AUC 0.989 on HC3 and 0.975 on TuringBench, but fails on RAID because RAID includes non-GPT families such as Llama, Mistral, MPT, and Cohere. Mistral token features are strong for contemporaneous modern generators (RAID and SemEval) but weak on older generator regimes. Model-free statistics are robust for RAID and Pile but nearly random on SemEval, where modern AI text has converged toward human surface statistics. This result suggests that detection features have clear shelf lives. GPT-2 perplexity should be understood as a reference-model similarity feature rather than a universal AI-generation signal. Similarly, Mistral token features are effective only when the observer model is close to the target generator era. Therefore, cross-dataset evaluation should not only compare final AUC, but also identify which feature groups remain valid under generator and domain shifts.

4.4 Noise Dilution

SemEval illustrates a counter-intuitive failure mode. Mistral token features alone reach AUC 0.976, but adding weak model-free features reduces performance to 0.833, and adding all 118 features yields only 0.847. This is not because token features become less informative; rather, XGBoost spends split capacity on many weak features that accidentally correlate with labels in the training set. The dilution is asymmetric: when the weak side has many features (88 weak features versus 30 strong token features), the harm is large; when the weak side is small, the harm is mild. This explains why SemEval suffers a large drop, while TuringBench remains almost unchanged. This supports the practical conclusion that feature fusion should be signal-aware rather than purely additive: weak feature groups should be screened out before they are concatenated with strong ones.

4.5 SHAP Explanations

SHAP analysis confirms that the RAID detector does not rely on a single brittle signal. Model-free features contribute most of the

global attribution, especially paragraph-level structure, while token-probability and embedding features provide complementary reference-model evidence.

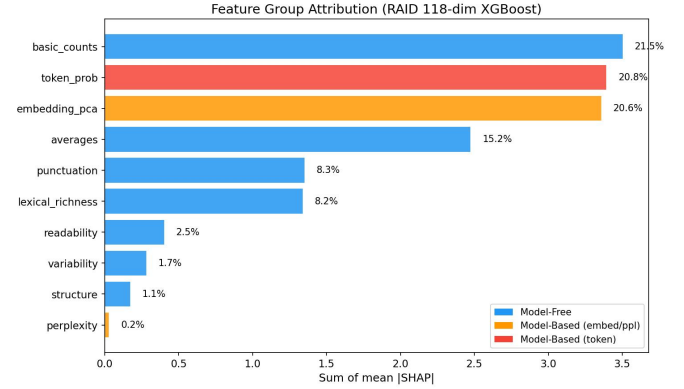


Figure 4: SHAP group attribution on RAID: model-free and reference-model features contribute complementary signal.

The most important individual signals are paragraph and sentence structure: words_per_paragraph, sentences_per_paragraph, paragraph_count, and sentence_length_std, as shown in the SHAP summary in Figure 5. This supports the qualitative observation that LLMs often produce organized, medium-length paragraphs with less rhythm variation than humans. These features are linguistically meaningful rather than arbitrary dataset identifiers. They suggest that LLM-generated text often has organized, medium-length paragraphs and lower rhythm variation, while human writing tends to show more irregular paragraph length and sentence structure. Therefore, the SHAP results support the claim that the detector is interpretable: it relies on structural and probability-based writing patterns instead of memorizing exact strings.

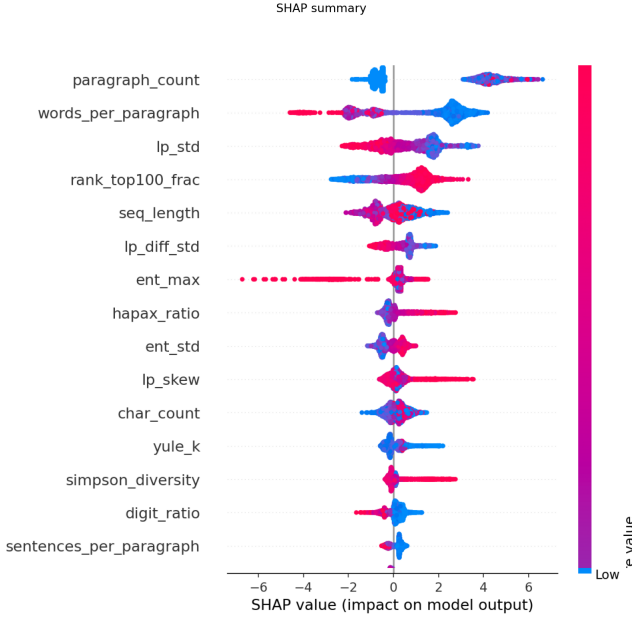
Beyond global importance, SHAP also explains individual decisions. Figure 6 is a single-document waterfall plot over the 118-feature model, decomposing one prediction from the base value $\mathbb{E}[f(x)]$ to the final output. Each bar is a feature’s signed contribution: red bars push the prediction toward the AI class and blue bars toward human. Model-free structural features (words_per_paragraph, paragraph_count) and observer-model token statistics (rank_top100_frac, lp_mean) appear interleaved among the top contributors, confirming at the instance level that the detector combines the two layers rather than relying on either alone. This per-prediction attribution is what makes the classifier auditable: every decision can be traced to specific, human-readable writing properties.

4.6 Adversarial Robustness

Table 5 summarizes attack robustness on RAID. Perturbations such as paragraph insertion, whitespace modification, synonym replacement, and homoglyph substitution cause only limited AUC drops. This is because the model mainly relies on aggregate linguistic statistics and paragraph-level patterns rather than exact string matching. This observation is consistent with the SHAP results in Section 4.5, where paragraph and sentence structure features contribute strongly to the final prediction. However, zero-width space and paraphrase attacks lead to much larger degradation. Zero-width spaces corrupt character- and whitespace-based statistics, while paraphrasing rewrites the text

Table 4: Cross-dataset ablation (XGBoost AUC). Rows are feature configurations; columns are datasets. The table shows that feature groups have different shelf lives.

Feature configuration	RAID	HC3	SemEval	TuringBench	Pile
model-free statistics	0.994	0.995	0.517	0.670	0.962
BERT embedding	0.774	0.877	0.498	0.539	0.895
GPT-2 perplexity	0.490	0.989	0.511	0.975	0.504
Mistral token probability	0.990	0.499	0.976	0.520	0.503
all reference-model features	0.994	0.994	0.961	0.970	0.894
model-free + BERT	0.995	0.996	0.516	0.736	0.973
model-free + token	0.999	0.994	0.833	0.672	0.961
all 118 features	0.999	1.000	0.847	0.985	0.973

**Figure 5: SHAP beeswarm summary of the top-15 features for the 118-feature RAID model.****Table 5: Adversarial attack robustness for the RAID detector.**

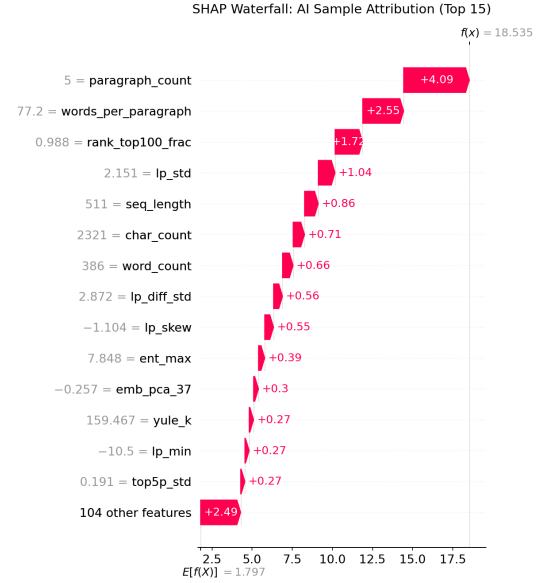
Attack	AUC	Drop
insert paragraphs	0.9915	-0.004
whitespace	0.9701	-0.025
synonym	0.9660	-0.029
homoglyph	0.9516	-0.044
zero-width space	0.8424	-0.153
paraphrase	0.7902	-0.205

at lexical, syntactic, and paragraph levels. Therefore, paraphrase directly changes the core structural signals used by the detector and becomes the most effective attack, reducing AUC by 0.205.

5 Discussion

5.1 Why Perplexity Fails

Perplexity is often described as a general AI-text signal, but our results show that it is better understood as *reference-model similarity*. GPT-2 perplexity detects text that is close to GPT-family distributions. It

**Figure 6: Instance-level SHAP waterfall decomposing one RAID prediction; red pushes toward AI, blue toward human.**

works on HC3 and TuringBench but fails on RAID, where generators span multiple model families. Thus, perplexity is not a universal detector of AI generation; it detects compatibility with a particular language model family.

5.2 Deployment Implications and Limitations

The practical lesson is to avoid one-size-fits-all detectors. For unknown or mixed generators, model-free statistics are a stable baseline. For modern LLMs, a contemporaneous observer model such as Mistral or Qwen can provide strong token-probability features. For older GPT-family text, GPT-2 perplexity can still be useful. Before deployment, a detector should estimate feature-group signal on a small labeled validation set and drop near-random groups to avoid noise dilution. Observer models should also be updated periodically as new generators appear.

These recommendations come with caveats. Our study focuses on English datasets and the available RAID attack set. Some feature computations require GPU inference for BERT, GPT-2, or Mistral, although model-free features are cheap. The results also depend on available labeled validation data for signal-aware feature selection.

Finally, paraphrase attacks remain challenging and require future semantic-level defenses.

6 Conclusion

We presented an interpretable AI-generated text detector built around a two-layer feature taxonomy. The 118-feature XGBoost model achieves AUC 0.9992 on RAID and remains explainable through SHAP. More importantly, the cross-dataset analysis shows that detection features have shelf lives: reference models only detect the generator regimes they align with, and model-free statistics weaken as modern AI writing approaches human surface statistics. Naive feature concatenation can hurt through noise dilution, so practical detectors should select feature groups according to target-domain signal rather than assuming more features are always better.

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